SCI-NOI v0.1

SCI-NOI v0.1 r0.3 Shared Normative Core

Implementation-blind Stage B draft

Scientific owner: Grant Wilson

SCI-NOI_NORMATIVE_CORE v0.1/draft-r0.3

Not frozen. No implementation conformity, validation, calibration, physical-noise, covariance-completeness, significance, performance, readiness, or production claim.

r0.18 packet b6f8e725...77ca0 | r0.3 owner 4b622de9...77e5

SCI-NOI v0.1 r0.3 Shared Normative Core

Document identity: SCI-NOI_NORMATIVE_CORE v0.1/draft-r0.3

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft; not frozen.

The complete shared authority is the following six files in fixed order:

1. normative/NOTATION.md
2. normative/DEFINITIONS.md
3. normative/EQUATIONS.md
4. normative/ASSUMPTIONS.md
5. normative/REQUIREMENTS.md
6. normative/PREDICTIONS.md

They are byte-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.3 in NORMATIVE_MODULE_BINDING.json, binding-file SHA-256 b7756dbeeace639e4ec86878466fb2afcc20fe5b8d256b48b6747bff1527e03c.

The binding also records SCI-NOI_OWNER_DECISION_SUPPLEMENT v0.1/r0.3, SHA-256 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5. That supplement approves the r0.2 architecture basis, base scope, and complete ODQ-102D design without modifying the immutable r0.18 packet.

Any module byte change requires a new binding and document generation. No rationale, ECS procedure, table, register, or PDF rendering supersedes these modules. The normative-core PDF appends their exact text in the stated order.

Authority and claim boundary

The r0.18 manifest and owner bytes remain immutable. Authorized SCI-VAL successors prove registration but author no NOI science. This draft establishes no implementation conformity, validation, calibration, physical-noise validity, covariance completeness, significance, performance, readiness, freeze, or production authorization.

Appendix A. Exact Bound Normative Modules

SCI-NOI v0.1 r0.3 Shared Normative Module: Notation

Module identity: SCI-NOI_NORMATIVE_NOTATION v0.1/r0.3

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft authority; content-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.3; not frozen.

- Symbol | Exact meaning
- i | One exact retained PTC occurrence with immutable observation, detector/channel, stable sample, segment, array/network, time, validity, ancestry, and provenance identity.
- d(i) | Stable observation-scoped detector/channel owning occurrence i.
- d | One stable realized detector/channel in one exact observation; the ordinary coherence unit.
- h(d) | Exact stable readout-network stratum of d; missing or ambiguous membership is unavailable for positive-mass detectors.
- b | One admitted realization-member identity in one exact design generation.
- p | One exact frozen MAP output row/pixel on its declared WCS, support, response, validity, and lifecycle domain.

- Z_i^PTC | Exact PTC transformed signal occurrence.
- gamma_i | Exact PTC-owner-selected MAP-facing coefficient, including family, generation, value, and QC state.
- G_pi | Exact frozen MAP projection/gridding contribution.
- a_pi | Exact frozen positive pre-normalization contribution G_pi gamma_i.
- C_p | Exact frozen set of admitted PTC occurrences contributing to row p.
- Q_p | Exact unsigned frozen MAP normalization sum_{i in C_p} a_pi; never signed or relearned by NOI.
- beta_d | Detector coefficient mass sum_p sum_{i in C_p,d(i)=d} a_pi.
- D_h^+ | Exact active design population {d : beta_d > 0 and exact h(d)=h}.
- epsilon_bd | Sign for active detector d in member b, in {-1,+1}. It is undefined for nonactive detectors.
- tau_h | Exact plan-bound balance tolerance for active stratum h, in [0,1), with no default.
- mathcal_D | Complete finite design identity: law or measure, population/order, alphabet, probabilities/weights, relations, mechanics, counts, ranks/null spaces, reconstruction, lifecycle, generation, completion, and failure.
- D_epsilon_b | Occurrence-domain diagonal operator obtained by expanding member b's active detector assignment over the exact admitted occurrence population.
- M_b(p) | NOI realization-map value produced by the exact frozen ordinary operator.
- m_MAP(p) | Independently governed immutable ordinary SCI-MAP signal for the corresponding all-+1 arithmetic, when scientifically realized.
- Vhat_cond(p) | Product conditional_detector_sign_randomization_marginal_second_moment on the exact common all-member domain.
- rho_cond(p) | Proposed reciprocal successor 1/Vhat_cond; unavailable pending owner decision and exact successor binding.
- sigma_cond(p) | Positive conditional scale sqrt(Vhat_cond(p)).
- zeta_cond(p) | Standardized product m_MAP(p)/sigma_cond(p), unit exactly 1.
- N_requested | Preregistered requested member count.
- N_resolved | Successfully resolved member count.
- N_completed | Members reaching complete producer terminal state.
- N_admitted | Members admitted to one exact named UNC use.
- N_unique_sign | Byte-identical-distinct canonical active assignment vectors.
- N_unique_orbit | Distinct exact complement orbits {epsilon,-epsilon}.
- r_sign | Uncentered active sign-matrix rank for the known-zero-center method.
- r_map | Exact rank, rank limit, or unavailable state after named map-domain projection.
- I_attempt | Execution-attempt/evidence identity; may change during an idempotent retry without changing scientific identity.

Inactive r0.1/r0.2 spellings appear only in immutable source quotations or the change map. They are not parallel active identifiers.

SCI-NOI v0.1 r0.3 Shared Normative Module: Definitions

Module identity: SCI-NOI_NORMATIVE_DEFINITIONS v0.1/r0.3

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft authority; content-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.3; not frozen.

D-001. Separate operations

NOI-GEN generates one declared finite ensemble. NOI-UNC infers one declared empirical quantity from one exact admitted ensemble. NOI-STD joins one immutable signal numerator to one authorized positive scale. No operation automatically realizes the next.

D-002. Ordinary route

NOI-GEN/PTC-TO-FROZEN-MAP-CONDITIONAL-SIGN@1 inserts the NOI assignment at the exact PTC-to-MAP boundary. MAP applies it inside otherwise frozen accumulation; NOI owns design, assignment, ensemble, and realization identity.

D-003. Fixed and relearned methods

A fixed method holds every consequential adjacent state at Theta0. A relearned method identifies every rerun/relearn stage and member state Theta_b. Different graphs never share an ensemble.

D-004. Complete and active populations

The complete population contains every stable detector/channel in one exact observation and TolTEC array. D_h^+ contains only exact positive-mass members of stratum h. Zero-mass, zero-total-stratum, singleton-active-stratum, and ambiguous-membership states are separately typed under the approved design.

D-005. Source-bearing conditional randomization ensemble

The fixed parent may contain astronomical source, structured residuals, and source-model error. Network-local coefficient balance may suppress some detector-common source contribution under additional assumptions, but supplies no pixelwise cancellation, source-free null, calibrated null, or repeated physical-noise ensemble.

D-006. Scientific replacement and execution retry

A changed assignment, plan, parent, operator, or scientific substitute is a new scientific identity. An infrastructure retry may retain scientific identity only for identical deterministic work with a new I_attempt.

D-007. Selected finite design

The approved design draws independent symmetric Rademacher candidates, accepts the first vector satisfying every active stratum's exact tau_h condition, and repeats with replacement under a positive preregistered cap. Conditional on resolution, its law is uniform over admitted vectors.

D-008. Initial UNC product

The sole primary name is conditional_detector_sign_randomization_marginal_second_moment. Because the selected target law has known mean zero, this marginal second moment equals the target-law marginal variance. It is not displayed interchangeably as "variance/second moment" and is never shortened to map variance, noise variance, physical-noise variance, or total uncertainty.

D-009. Common all-member domain

Initial UNC uses exactly the rows where every admitted member supplies a valid finite value. Survivor, pairwise, zero-filled, interpolated, and extended domains are different unsupplied methods.

D-010. Counts, rank, and information

Member counts, sign uniqueness, complement-orbit uniqueness, r_sign, r_map, effective Monte Carlo information, and estimator uncertainty are separate facts. A pointwise moment, retained ensemble, covariance, reciprocal, precision, and consumer weight are distinct product roles.

D-011. Reciprocal disposition

The r0.18 reciprocal identity is immutable history. r0.3 neither places a reciprocal in the ordinary base bundle nor claims a renamed successor. Every reciprocal, inverse variance, precision, and consumer weight remains unavailable pending exact owner disposition and applicable profile/Registry binding. STD consumes sqrt(Vhat_cond) directly.

D-012. Standardized product

zeta_cond is the independently realized immutable ordinary MAP signal divided by sqrt(Vhat_cond) on their exact compatible finite-positive domain. Its unit is 1; it is not significance, probability, detection, or catalog authority.

D-013. Persistence, profiles, and ownership

Persistence is plan-selected. External transforms remain externally owned. GEN owns completion facts; NOI owns named-use policy; SCI-VAL binds/evaluates but authors neither. Existing r0.18 profile/Registry objects are immutable; changed r0.3 bytes require exact successors before evaluation.

SCI-NOI v0.1 r0.3 Shared Normative Module: Equations

Module identity: SCI-NOI_NORMATIVE_EQUATIONS v0.1/r0.3

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft authority; content-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.3; not frozen.

SCI-NOI-EQ-001 - frozen contribution

```
a_pi = G_pi gamma_i.
```

SCI-NOI-EQ-002 - unsigned normalization

```
Q_p = sum_{i in C_p} a_pi.
```

SCI-NOI-EQ-003 - detector mass and active population

```
beta_d = sum_p sum_{i in C_p, d(i)=d} a_pi,
D_h^+ = {d : beta_d > 0 and exact h(d)=h}.
```

SCI-NOI-EQ-004 - exact active-stratum condition

```
Delta_h(epsilon_b)
  = abs(sum_{d in D_h^+} epsilon_bd beta_d)
    / sum_{d in D_h^+} beta_d
  <= tau_h,                 0 <= tau_h < 1.
```

The ratio is evaluated only for an active positive-total stratum. tau_h is exact, plan-bound, and has no default.

SCI-NOI-EQ-005 - ordinary realization member

```
M_b(p)
  = [sum_{i in C_p} a_pi epsilon_b,d(i) Z_i^PTC] / Q_p.
```

The sign modifies only Z_i^PTC, exactly once. All coefficients, membership, normalization, WCS, projection, support, coverage_cut, response handling, validity, and MAP-local gates remain frozen.

SCI-NOI-EQ-006 - independently governed ordinary MAP signal

```
m_MAP(p) = [sum_{i in C_p} a_pi Z_i^PTC] / Q_p.
```

This is the corresponding all-+1 arithmetic, not an NOI-generated product. m_MAP exists only when the independently governed ordinary SCI-MAP route is scientifically realized. Missing coefficient, numerical coverage_cut, MAP admission, or other required parent makes m_MAP and dependent STD unavailable.

SCI-NOI-EQ-007 - fixed signal operator

```
R_b^fixed = A_MAP,Pi D_epsilon_b H_PTC^fixed.
```

D_epsilon_b is the occurrence-domain diagonal expansion of the active detector assignment over the exact admitted occurrence population. Required response companions use identical occurrence membership and the identical modifier exactly once; missing response is unavailable and never shrinks the signal population.

SCI-NOI-EQ-008 - active-domain expectations

```
E_D[epsilon_bd] = 0, for d in the exact active design population,
E_D[M_b(p) | fixed parent and operator] = 0,
  for p in the exact available frozen MAP row domain.
```

The relations require the approved complement-symmetric target law. No sign is implied for nonactive detectors.

SCI-NOI-EQ-009 - conditional marginal second moment

```
Vhat_cond(p) = sum_{b in A_UNC} [1/N_resolved] M_b(p)^2.
```

The known target center is zero; no ensemble-mean subtraction or B-1 rule applies. The equation uses the exact common all-member domain.

SCI-NOI-EQ-010 - unavailable proposed reciprocal

```
rho_cond(p) = 1 / Vhat_cond(p).
```

This equation defines no active base product. It remains unavailable pending owner disposition and exact profile/source binding.

SCI-NOI-EQ-011 - conditional scale

```
sigma_cond(p) = sqrt(Vhat_cond(p)).
```

SCI-NOI-EQ-012 - standardized MAP signal

```
zeta_cond(p) = m_MAP(p) / sigma_cond(p).
```

The output exists only on the exact compatible finite-positive domain, has unit 1, and carries only the claim in SCI-NOI-REQ-031.

SCI-NOI v0.1 r0.3 Shared Normative Module: Assumptions

Module identity: SCI-NOI_NORMATIVE_ASSUMPTIONS v0.1/r0.3

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft authority; content-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.3; not frozen.

SCI-NOI-ASM-001 - exact authority

Scientific authority is the 17 objects in SCI-NOI_AUTHOR_PACKET_MANIFEST v0.1/r0.18, SHA-256 b6f8e7252e7f61f4506899cb3e8e26cf939887bb48464852713f8ce81ac77ca0, its controlling owner object SHA-256 272ac939b8a7109a123073b1a39fcdd7ac4129c603683ee81257b94ab2f55a0b, and SCI-NOI_OWNER_DECISION_SUPPLEMENT v0.1/r0.3, SHA-256 4b622de96e72860e544bf3699e3a131bc9cca5d2be0b0fe5bb41096327e977e5. Authorized SCI-VAL records supply binding evidence only.

SCI-NOI-ASM-002 - frozen numerical gates

The ordinary numerical route remains unavailable until every exact PTC coefficient/QC, numerical coverage_cut, MAP admission, and parent fact exists.

SCI-NOI-ASM-003 - fixed operator

Occurrences, coefficients, projection, denominator, WCS, support, boundary, response, validity, and MAP-local gates are identical for the real signal and every member. Reselection or relearning creates another method.

SCI-NOI-ASM-004 - approved scope and design

Base v0.1 has one independent ensemble per exact observation and TolTEC array. The owner approved the complete bounded first-accepted rejection design in the r0.3 supplement; no cross-observation randomization or coadd is supplied.

SCI-NOI-ASM-005 - conditional uniformity

Independent symmetric candidates have equal base probability. First acceptance under a member-independent admissibility set and finite cap is uniform over admitted vectors conditional on successful resolution. The cap changes resolution probability, not this conditional law.

SCI-NOI-ASM-006 - conditional target only

The fixed parent may contain source and structured residuals. The target law is not repeated physical noise, total MAP uncertainty, a source-free null, or a calibrated tail law.

SCI-NOI-ASM-007 - atomic completion

Candidate rejection is not member failure. Once admitted, failure of any member fails the ensemble for base UNC; diagnostics are not survivor inference.

SCI-NOI-ASM-008 - no significance model

No Gaussian, Student, pivotal, random-field, selection, multiplicity, false-alarm, completeness, purity, or catalog model is supplied.

SCI-NOI-ASM-009 - immutable successors

Adjacent parents and prior products remain immutable. Transformation, Wiener, and FRUIT learning/selection/replay require exact external authority and new generations.

SCI-NOI-ASM-010 - profile revision boundary

The four exact r0.18 profile bytes and Registry objects remain immutable. Changed r0.3 notation/actions have no evaluable successor until exact owner approval and Registry/source binding.

SCI-NOI v0.1 r0.3 Shared Normative Module: Requirements

Module identity: SCI-NOI_NORMATIVE_REQUIREMENTS v0.1/r0.3

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft authority; content-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.3; not frozen.

shall is required; shall not is prohibited; not_applicable and unavailable are typed scientific states, not numerical sentinels.

SCI-NOI-REQ-001 GEN, UNC, and STD shall remain separate operations. None shall automatically realize, authorize, or mutate another.

SCI-NOI-REQ-002 The ordinary route shall be NOI-GEN/PTC-TO-FROZEN-MAP-CONDITIONAL-SIGN@1; signed outputs shall be NOI realizations, not ordinary MAP science products.

SCI-NOI-REQ-003 Every GEN method shall classify consequential adjacent state as fixed, rerun/relearned, not applicable, or unavailable. Different graphs shall not share an ensemble.

SCI-NOI-REQ-004 The coherence unit shall be one stable detector/channel in one exact observation. Its sign shall be constant across all admitted occurrences and ordered by canonical scientific identity.

SCI-NOI-REQ-005 Network balance shall use frozen positive coefficient mass beta_d. No network, array, or observation shall balance another, and beta_d shall not be promoted to another scientific role.

SCI-NOI-REQ-006 Base v0.1 shall produce one independent observation-level ensemble per exact observation and TolTEC array. It shall define no cross- observation randomized ensemble, randomized coadd, or multi-observation GEN route. Any future route requires the separate contract listed in the r0.3 owner supplement.

SCI-NOI-REQ-007 Every ordinary member shall satisfy SCI-NOI-EQ-005. The sign shall act only on Z_i^PTC, exactly once; all other MAP facts and gates shall remain frozen.

SCI-NOI-REQ-008 Inline and materialized application shall be equivalent only for identical equation, occurrence membership, assignment, frozen operator, and output domain. All-+1 arithmetic shall not manufacture an SCI-MAP product.

SCI-NOI-REQ-009 SCI-NOI-EQ-007 shall govern the fixed signal/response relation. Required response companions shall use identical membership and the modifier exactly once; missing response shall be unavailable without hidden signal subsetting.

SCI-NOI-REQ-010 The complete population, D_h^+, zero-mass detectors, zero-total strata, singleton active strata, and ambiguous positive-mass membership shall remain separately typed. No zero-denominator ratio shall be evaluated.

SCI-NOI-REQ-011 Zero-mass detectors shall be excluded from the active sign matrix but retained as typed facts. Zero-total strata shall be not_applicable. Ambiguous positive-mass membership and every singleton active stratum with tau_h < 1 shall make design resolution unavailable.

SCI-NOI-REQ-012 The selected ODQ-102D design shall use exact tau_h in [0,1) without default, independent symmetric Rademacher candidates, first-accepted bounded rejection, a positive exact preregistered cap, fail- closed exhaustion, with-replacement draws, equal weight 1/N_resolved, scheduling-independent content-bound generator/keying, and no tolerance relaxation or best-failed fallback.

SCI-NOI-REQ-013 Every design shall bind an exact target probability law or deterministic finite measure; stable population/order; assignment alphabet; probabilities/weights; equality, duplicate, complement, and equivalence relations; requested/resolved/completed/admitted/unique counts; applicable ranks/null spaces; deterministic reconstruction identity; lifecycle, generation, completion, and failure. Every applicable field shall be preregistered before candidate or member output inspection. Unused fields shall be not_applicable.

SCI-NOI-REQ-014 RNG identity, seed/key bytes, tolerance, retry cap, and rejected-candidate count shall be required only for a selected stochastic rejection family. For the selected family they shall be exact and plan-bound. Conditional on success, finite first-accepted rejection shall be uniform over the admitted vectors; the cap shall change resolution probability only.

SCI-NOI-REQ-015 Under every accepted design, exact duplicate equality and exact complement-orbit relation shall be represented separately. Under the owner-selected independent with-replacement rejection family, complement- related assignments shall remain distinct and no complement pair shall be forced.

SCI-NOI-REQ-016 Exact duplicates shall remain distinct draws. Products shall report all six N_* counts, r_sign, r_map, and use-specific effective information separately; none shall substitute for another.

SCI-NOI-REQ-017 The source-bearing conditional randomization ensemble shall carry D-005's objective and exclusions. Zero target-law mean shall not imply source removal from members, moments, or tails.

SCI-NOI-REQ-018 Every admitted assignment shall complete through the exact frozen operator. Any admitted-member failure shall fail the initial UNC ensemble; survivors and partial accumulators are diagnostic only.

SCI-NOI-REQ-019 An idempotent retry may retain scientific identity only for identical deterministic work with a new I_attempt. Changed scientific state shall be a replacement with new identity.

SCI-NOI-REQ-020 Each persistence plan shall select persisted, compact-regeneration, streaming-sufficient-statistic, or a separately identified composite and shall declare reconstruction, regeneration, sufficiency, unavailable later questions, lifecycle, completion, and failure.

SCI-NOI-REQ-021 An external deterministic transform shall require exact owner authority and parity for every operator fact; NOI shall not choose, tune, relocate, simplify, substitute, or silently omit it.

SCI-NOI-REQ-022 Wiener or FRUIT learning, selection, continuation, update, or per-member replay shall create every applicable new immutable owner, product, GEN, and UNC generation. Base numerical transformed routes remain unavailable.

SCI-NOI-REQ-023 Every UNC method shall bind exact target law, GEN membership, center, estimator, weights, dependence/missingness, domain, WCS/support/ response/unit, representation, ranks/null spaces, regularization, omissions, estimator uncertainty, lifecycle, claim, and named use.

SCI-NOI-REQ-024 Initial UNC shall consume only an exact all-members- successful ensemble admitted by the named profiles. Rejected candidates, failed ensembles, survivors, pairwise subsets, mixed methods, failed persistence, and partial streaming state shall not be admitted.

SCI-NOI-REQ-025 Initial UNC shall compute SCI-NOI-EQ-009 on the exact common domain. It shall use the known zero target center and shall not subtract the finite ensemble mean or apply B-1.

SCI-NOI-REQ-026 The primary product name shall be conditional_detector_sign_randomization_marginal_second_moment, with squared signal units. The equality to target-law marginal variance shall be stated separately; prohibited shortened names and physical-noise/total-uncertainty/ covariance/precision/significance claims shall not be used.

SCI-NOI-REQ-027 Numerical estimator uncertainty shall be explicit or unavailable. Counts, uniqueness, ranks, and effective information shall remain separate from that uncertainty.

SCI-NOI-REQ-028 Retained ensemble, marginal moment, projection, structured/full covariance, reciprocal, precision, weight, and unavailable representations shall remain separate roles. Missing covariance is unknown, not zero or independence.

SCI-NOI-REQ-029 No reciprocal shall enter the ordinary base bundle without a new exact owner disposition. Every reciprocal, inverse variance, precision, and consumer weight shall remain unavailable; STD shall consume sqrt(Vhat_cond) directly. Any future reciprocal shall use the exact name reciprocal_conditional_randomization_second_moment, finite-positive domain, separate identity/profile/use, and no prohibited promotion.

SCI-NOI-REQ-030 Initial STD shall use independently realized m_MAP, sigma_cond, and zeta_cond on the exact compatible finite-positive intersection. Missing MAP coefficient, numerical coverage_cut, MAP admission, or other required parent shall make m_MAP and STD unavailable. JINC remains separate and unavailable.

SCI-NOI-REQ-031 zeta_cond shall have unit 1 and the sole claim: "MAP signal standardized by the stated conditional detector-sign-randomization second-moment scale." No significance, probability, detection, false-alarm, completeness, purity, or catalog claim shall follow.

SCI-NOI-REQ-032 Admission shall retain separately named request, applicability, eligibility, and realization fields. Only the exact positive conjunction shall project to the exact named action.

SCI-NOI-REQ-033 Generation-input admission shall admit only one exact candidate occurrence for the selected GEN method; later operations remain separate.

SCI-NOI-REQ-034 UNC-member admission shall consume but not redefine GEN completion, failure, duplicate/equivalence, support, imprint, QC, persistence, lifecycle, cause, and provenance facts.

SCI-NOI-REQ-035 UNC-ensemble admission shall admit only one exact complete all-members-successful ensemble for one estimator/domain and shall not realize the estimator or authorize another use.

SCI-NOI-REQ-036 STD admission shall permit only the exact initial numerator/ scale join and claim. r0.3 evaluation shall remain unavailable until exact successor profile/Registry approval and binding.

SCI-NOI-REQ-037 Missing, conflicting, incomplete, unsupported, invalid, or failed required facts shall yield typed states. No undocumented sentinel, filename, shape, approximate WCS, or hidden default shall establish success.

SCI-NOI-REQ-038 PTC, MAP, JINC, transform, Wiener, FRUIT, and prior NOI parents shall remain immutable; later results shall be new versioned companions with exact ancestry and lifecycle.

SCI-NOI-REQ-039 No assignment, uncertainty, reciprocal, precision, scale, or weight shall become validity, support, exposure, or a PTC/MAP coefficient without separately approved successor authority.

SCI-NOI-REQ-040 Existing r0.18 profile and Registry objects shall remain immutable. A proposed name shall not create an r0.3 successor. Affected r0.3 evaluation shall remain unavailable until exact successor approval and Registry/source binding.

SCI-NOI-REQ-041 The rationale and ECS shall bind the exact six shared modules and shall not independently author equations, requirements, or predictions. Any module byte change requires a new binding and document generation.

SCI-NOI-REQ-042 This draft shall establish no implementation conformity, numerical validation, calibration, physical-noise validity, covariance completeness, Gaussian significance, achieved performance, readiness, freeze, production suitability, or production authorization.

SCI-NOI v0.1 r0.3 Shared Normative Module: Predictions

Module identity: SCI-NOI_NORMATIVE_PREDICTIONS v0.1/r0.3

Scientific owner: Grant Wilson

Status: implementation-blind Stage B draft authority; content-bound by SCI-NOI_NORMATIVE_MODULE_BINDING v0.1/r0.3; not frozen. These are analytic expectations, not implementation-conformity or empirical-validation reports.

SCI-NOI-PRED-001 A globally complemented active assignment has equal target probability under the approved complement-symmetric law.

SCI-NOI-PRED-002 Every active detector has target-law sign mean zero; the fixed linear member has target-law mean zero on each exact available row.

SCI-NOI-PRED-003 Complement symmetry does not remove source or structured residual contributions from individual members, second moments, or tails.

SCI-NOI-PRED-004 Complementing every active sign negates M_b while leaving the unsigned Q_p unchanged.

SCI-NOI-PRED-005 Complement-related members have identical pointwise squares and outer products while remaining distinct draws; exact duplicate and orbit relations remain separately represented.

SCI-NOI-PRED-006 Vhat_cond is nonnegative wherever available and is unavailable outside the exact common all-member domain.

SCI-NOI-PRED-007 Source-bearing fixed structure can appear in Vhat_cond; zero target-law mean does not prevent it.

SCI-NOI-PRED-008 Any failed admitted member makes initial UNC unavailable; completed survivors do not realize a partial estimator.

SCI-NOI-PRED-009 More members may improve estimation of the declared randomization target but cannot create exposure or reduce the immutable parent's realized noise as 1/sqrt(N).

SCI-NOI-PRED-010 A fixed owner-defined linear transform propagates a conditional moment under its exact operator; a learned per-member transform does not satisfy that identity by implication.

SCI-NOI-PRED-011 A positive profile decision neither realizes the next operation nor changes producer-owned facts.

SCI-NOI-PRED-012 Missing required response makes response-dependent output unavailable; signal membership does not silently shrink.

SCI-NOI-PRED-013 An idempotent retry may change attempt/evidence identity without changing scientific identity; a changed assignment or scientific state cannot.

SCI-NOI-PRED-014 Conditional on successful resolution, first-accepted finite rejection from independent symmetric candidates is uniform over admitted sign vectors: every admitted vector has the same base probability, and the common geometric probability of earlier rejection cancels in the conditional ratio. The cap changes success probability, not the accepted-vector law.

SCI-NOI-PRED-015 A zero-total-mass stratum is not_applicable to balancing and produces no imbalance ratio.

SCI-NOI-PRED-016 A singleton active stratum makes the selected balance design unavailable for every tau_h < 1; it is neither exempted nor cross-balanced.

SCI-NOI-PRED-017 zeta_cond has unit 1 on its available domain; dimensionlessness establishes no pivotal, significance, probability, detection, false-alarm, completeness, purity, or catalog interpretation.